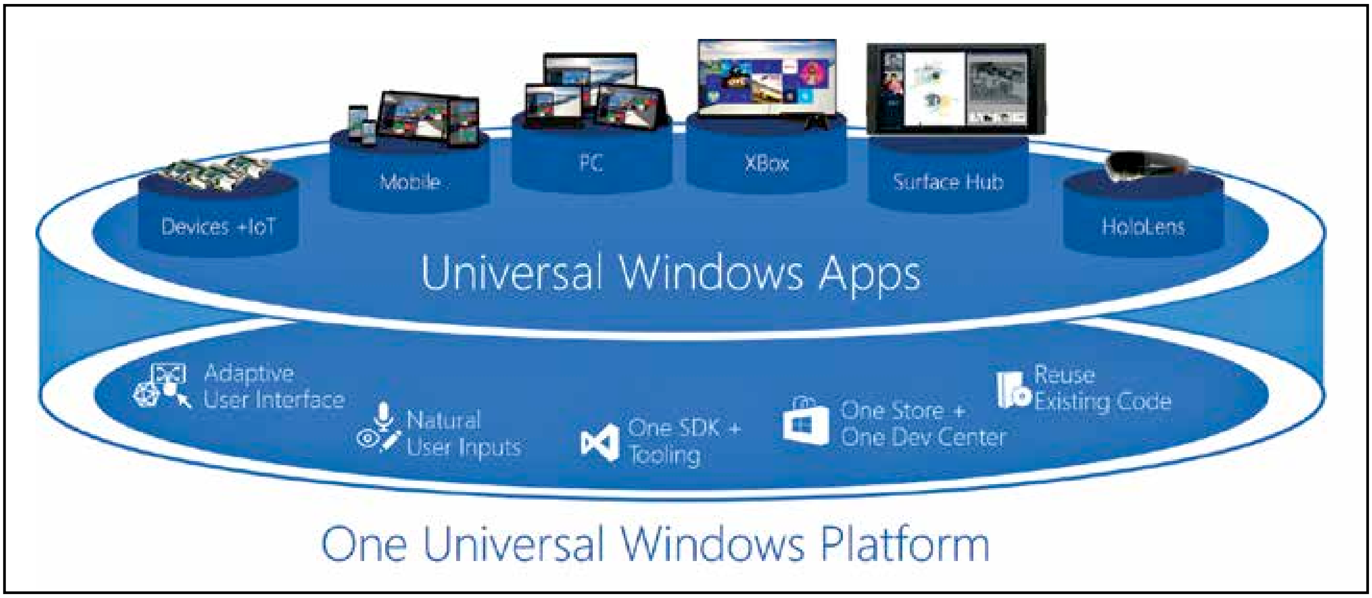


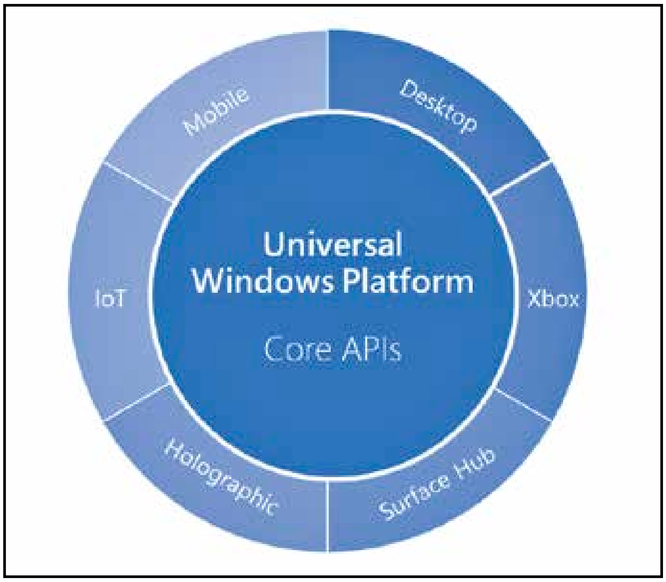
Universal Windows Apps can run on any Windows platform and use techniques such as adaptive UIs to run on different form factors and with different kinds of interactions patterns.

Before Windows 8, the mobile, desktop and the Xbox's OS stacks were very different. Windows for desktops used the Windows NT kernel, phones used the CE kernel and the Xbox used its own special kernel. Windows Phone 8 finally switched to the NT kernel making it a lot similar to Windows for desktops. Xbox One further converged these platforms by switching to the same kernel as Windows 8.

The application platforms and APIs above the kernel level were of course still quite different across platforms. While converging these kernels might have made things easier for Microsoft, it didn't benefit other developers as much.

With Windows 10 comes a true convergence of all devices to the same Windows kernel, 'universal hardware drivers', and 'standard network and I/O'. This in essence is what Windows Core is.

Not only has Windows Core expanded to all devices that already supported Windows, but the breadth of devices that support Windows has also expanded. For instance, other than PCs, tablets, and phones, the Xbox One will also soon update to use the Window 10 core, it is already available for the Raspberry Pi 2, and of course Microsoft's new HoloLens also runs Windows 10 core.



The new model for Windows APIs tries to as many APIs in common as possible. The only APIs missing should be those that are of no value on other platforms.

Consolidation and convergence are huge themes with Windows 10. After unifying the Windows core across platforms Microsoft has also unified the development SDKs across platforms into what they call the Universal Windows Platform (UWP).

Universal Windows Platform

UWP is a layer of APIs that are guaranteed by Microsoft to be available on any device that is running Windows, whether it be a mobile phone, a PC, the Xbox, an embedded device like the Raspberry Pi or even a device that hasn't been released yet.

There are still platform-specific features, that can be layered on top of the UWP via extension SDKs. These SDKs extend the feature set of Windows for a particular family of devices. For example, the mobile family of Windows might power phones and phablets, while the desktop family on Windows will power tablets, laptops and desktops. They both have their own extension SDKs that can be used to implement platform-specific features. Microsoft estimates though that over 85% of Windows APIs are available universally.

The same application can now support multiple device families more easily, not just by using only the common UWP APIs, but by detecting and using only the features available on that platform. There is now an API to check for the presence / absence of platform-specific features and APIs.

This is different from earlier versions when such checks could only be performed at compile time in order to build different binaries for different platforms. Now runtime checks allow developers to have the same application code adapt to mobile or desktop.

Consider that Windows 10 now makes it possible to have a phone power a full desktop experience if docked to a station that has a larger screen, a keyboard and mouse. Of course this will need a very powerful phone, and might not be feasible today, but the possibility of such a thing is eminent. Microsoft's new modular approach means that the OS for such a phone could simply be a composition of the